

Table of Contents

- [ProtocolDefinitionEditorDefTopic](#)
- [ProtocolDefinitionEditorUsing.2.01](#)
- [ProtocolDefinitionEditorUsing.2.02](#)
- [ProtocolDefinitionEditorUsing.2.03](#)
- [ProtocolDefinitionEditorUsing.2.04](#)
- [ProtocolDefinitionEditorUsing.2.05](#)
- [ProtocolDefinitionEditorUsing.2.06](#)
- [ProtocolDefinitionEditorUsing.2.07](#)
- [ProtocolDefinitionEditorUsing.2.08](#)
- [ProtocolDefinitionEditorUsing.2.09](#)
- [ProtocolDefinitionEditorUsing.2.10](#)
- [ProtocolDefinitionEditorUsing.2.11](#)
- [ProtocolDefinitionEditorUsing.2.12](#)
- [ProtocolDefinitionEditorUsing.2.13](#)
- [ProtocolDefinitionEditorUsing.2.14](#)

ProtocolDefinitionEditorDefTopic

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About Protocol Definition Editor

[About Protocol Definition Editor](#)

The Protocol Definition Editor provides a graphical view of protocol type definition files.

A protocol type definition file describes how to use protocol engines to mimic a protocol. The file type must be a valid Protocol Aware type definition file (.pa or .xml) or the name of a static protocol.

You can create protocol definitions using any XML or text editor or using the Protocol Definition Editor.

Currently, only nWire protocol definition files are supported. IG-XL supplies definition files for the other protocol types. Only nWire lets you create and customize files.

One use of the Protocol Definition Editor is for debugging. You can modify a frame or insert new cycles into a frame and see the effects.

The following topics are available:

- [Using the Protocol Definition Editor](#)
- [Port Debug Display](#)
- [Protocol Studio Toolbar and Command Reference](#)

IG-XL includes additional Protocol Aware tools:

- [Protocol Studio:](#)
 - [Displays Protocol Aware \(PA\) frames at various levels of abstraction](#)
 - [Enables programming of PA port settings](#)
 - [Enables initiation of frames](#)
- [CyclePro:](#)
 - [Raw cycle display mode](#)
 - [Symbol10b display mode](#)
 - [PCIe Link traffic decode \(CMEM\) and ATE Tx traffic decode \(Tx HRAM\)](#)

–	Layer Selection: Transaction, Data Link, PHY
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For more information, see:

- | | |
|---|---------------------------------------|
| • | About Protocol Studio |
|---|---------------------------------------|
- | | |
|---|--------------------------------|
| • | About CyclePro |
|---|--------------------------------|

For information about Protocol Aware in general and nWire in particular, see the following:

- | | |
|---|--------------------------------------|
| • | About Protocol Aware |
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- | | |
|---|--------------------------------------|
| • | nWire Protocol Aware |
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Using the Protocol Definition Editor

Using the Protocol Definition Editor

This section describes how to use the Protocol Definition Editor. It covers these topics:

- [Protocol Definition Editor Window](#)
- [Launching the Protocol Definition Editor](#)
- [Creating, Opening, Saving, and Closing Files](#)
- [Protocol Definition Window](#)
- [Frame Window](#)
- [Editing a Definition](#)
- [Inserting Elements](#)
- [Editor Toolbar and Command Reference](#)

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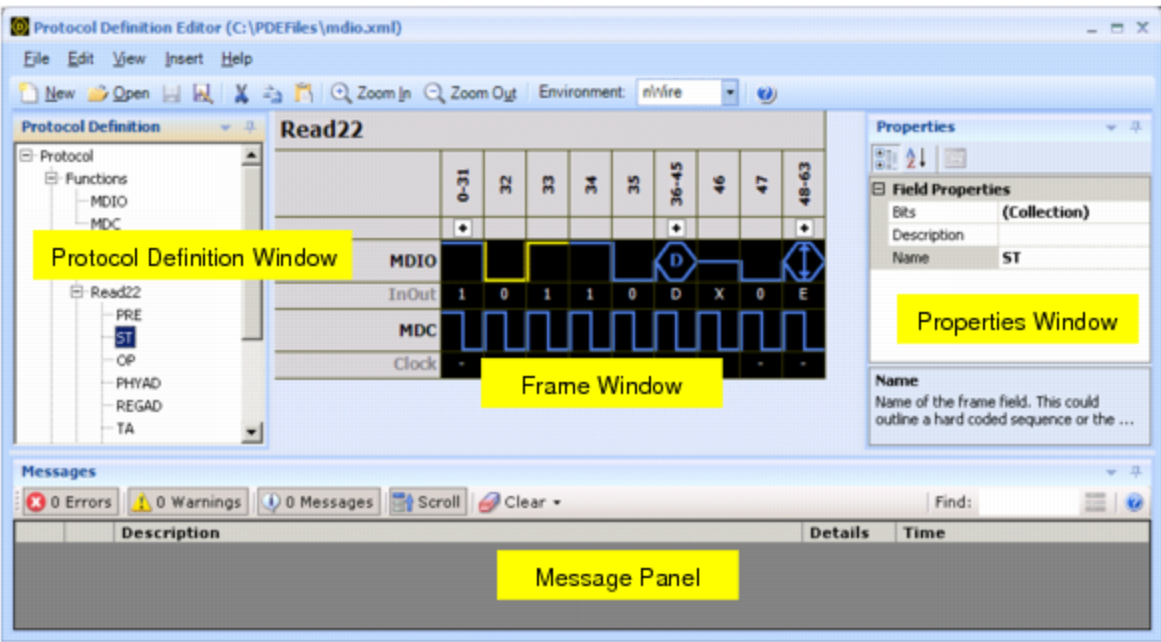
ProtocolDefinitionEditorUsing.2.02



Using the Protocol Definition Editor : Protocol Definition Editor Window

Protocol Definition Editor Window

When data has been loaded and a frame has been selected, the Protocol Definition Editor window looks like this:



Protocol Definition Editor Window

The parts of the window are as follows:

- The Protocol Definition window is a tree structure that shows the components of a definition. Select a component to display its properties in the Properties window. Select a frame or field to display the frame graphically in the Frame window. See Protocol Definition Window.
- The Frame window displays the currently selected frame. You can edit the pin's bit action. See Frame Window.

- [The Properties window displays the properties of the currently selected component. You can edit the property values.](#)
- [The Message Panel displays errors, warnings, and informational messages from the session. For help on this panel, see \[About the Message Panel\]\(#\).](#) To enable the Auto-Hide feature, so that the Message Panel appears only when a new message arrives, unpin the panel by clicking the pin icon.

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Using the Protocol Definition Editor : Launching the Protocol Definition Editor

Launching the Protocol Definition Editor

The Protocol Definition Editor runs independently of the rest of IG-XL and is completely stand-alone. It does not matter whether IG-XL is running. All its features are available whether or not IG-XL is running.

Launch the Editor from the Start menu. It is under the IG-XL Tools menu:

Start>All Programs>Teradyne IG-XL Vversion_uflx>Tools>Protocol Definition Editor



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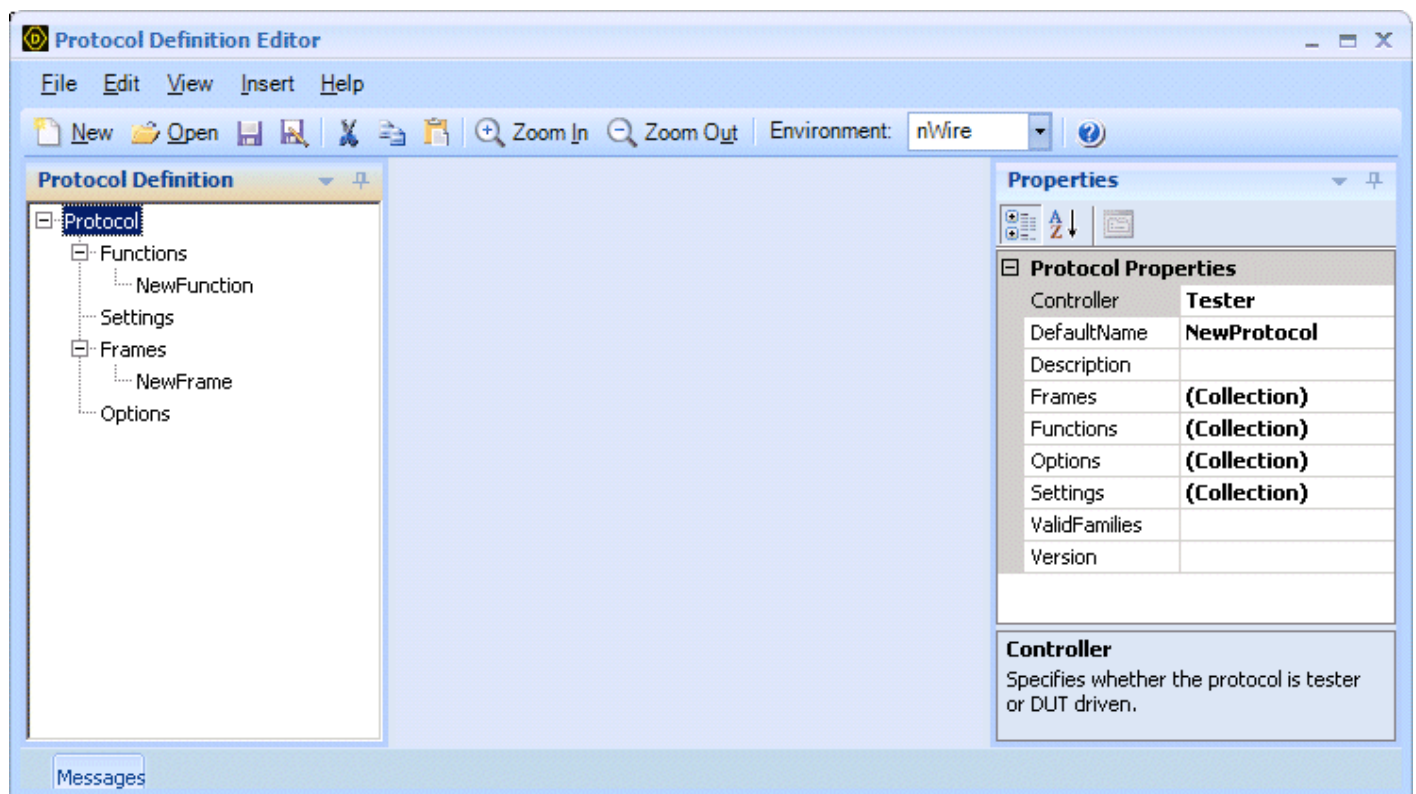


Using the Protocol Definition Editor : Creating, Opening, Saving, and Closing Files

Creating, Opening, Saving, and Closing Files

 **New** To create a new definition file, click this button or use the File>New command or Ctrl+N.

The window looks like this:



Protocol Definition Editor Window, Creating a New Definition File

You can insert new elements to populate the definition file. See [Inserting Elements](#).

 **Open** To open an existing definition file, click this button or use the File>Open command or Ctrl+O. A browser lets you navigate to the file. The browser finds files with the extensions .pa or .xml.

The title bar displays the path and filename of the opened file.

If problems occur opening the file, an error is written to the Message Panel. In particular, if you try to open an XML protocol definition file that contains a syntax error, you see this message:
Unable to read the file. Is this is a valid protocol definition file?
There is an error in XML document (26, 4).
The numbers indicate the line and character position in the file at which the error was discovered. This may not be where the error occurs, but it is where the Open function realized that an error occurred because the syntax does not make sense.
Once you edit a new or existing file, you can save it:

-  To save the changes to a file you opened, use this button, File>Save, or Ctrl+S. If this is a new file that has not yet been saved, a Save As is performed.
-  To save the changes to a new file, use this button or File>Save As. A browser lets you specify the location and filename.

If you have a file open when you use Open or New, the open file is closed before the new file is opened. If you have made changes to the file, you are asked if you want to save them.

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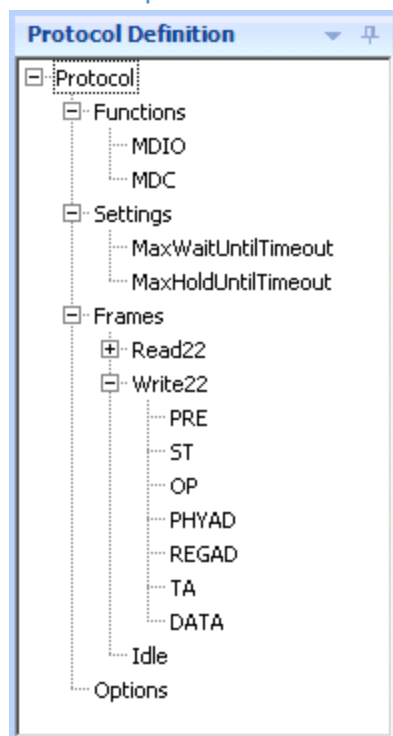


Using the Protocol Definition Editor : Protocol Definition Window

Protocol Definition Window

The Protocol Definition window displays the components of a protocol definition file in a tree structure. The structure's hierarchy corresponds to the XML schema for protocol type definition files. For more information, see [XML Schema for Protocol Type Definition Files](#) in the Protocol Aware section.

A loaded protocol definition file looks like this:



Protocol Definition File Structure

Briefly, the structure is as follows:

- Under Protocol are Functions, Settings, Frames, and Options.

- [Under](#) Functions are the individual named functions.
- [Under](#) Settings are the individual named settings, if any.
- [Under](#) Frames are the individual named frames, such as Read22. Under each frame are the individual named fields, such as PRE.
- [Under](#) Options are the individual named options, if any.

When you select an item in the Protocol Definition tree, its properties are displayed in the Properties window, where you can view and edit them.

If you select a frame or a field of a frame, the frame is displayed in the Frame window. If you select a field, it is highlighted in the Frame window. See [Frame Window](#).

On editing the selected item, see [Editing a Definition](#).

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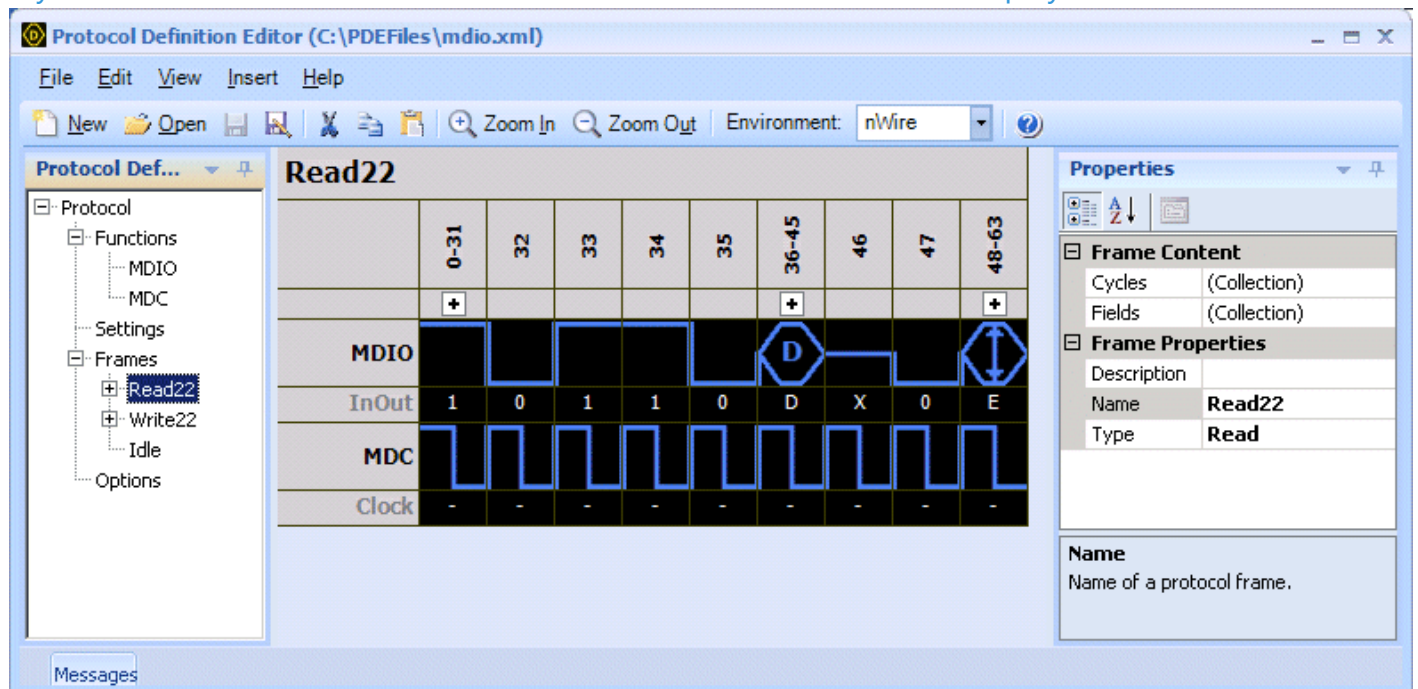
ProtocolDefinitionEditorUsing.2.06



Using the Protocol Definition Editor : Frame Window

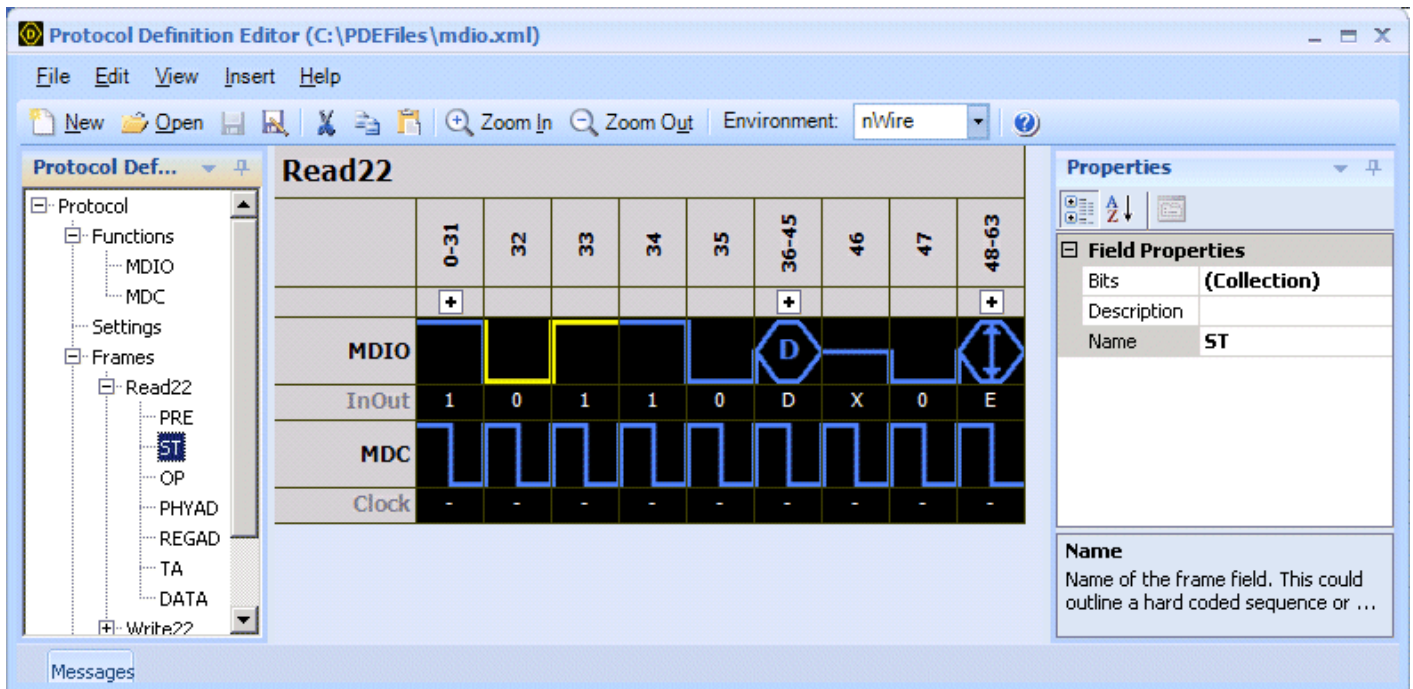
Frame Window

If you select a frame in the Protocol Definition window, the frame is displayed in the Frame window.



Protocol Definition Editor Window, Frame Selected

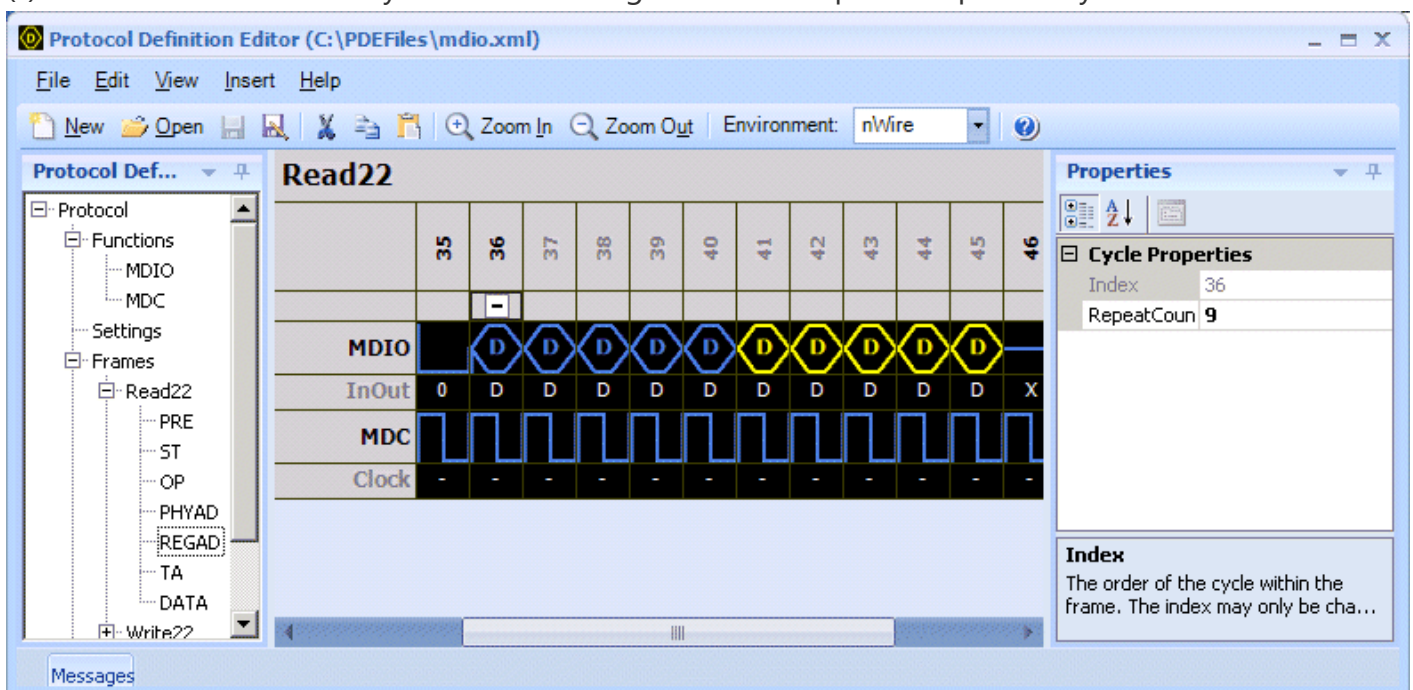
If you select one field of a frame, the bit actions corresponding to the field are highlighted in yellow.



Protocol Definition Editor Window, Field of a Frame Selected

The Frame window displays the bit actions associated with a frame. The grid has the cycle numbers along the horizontal axis and the functions along the vertical axis. Each function consists of two rows: the top has the function name and the graphical representation of the bit action, and the bottom has the function type and the characters that specify the bit action.

The column header shows the span of the cycles for which the bit action is valid. Cycles that repeat are represented as a grouped column. In the figure above, the repeated cycles are 0-31, 36-45, and 48-63. You can expose and hide the repeated cycles using the plus and minus control in the cell below the cycle numbers. Click the plus sign (+) to expose the repeated cycles and the minus sign (-) to hide all but the first cycle. The following shows the exposed repeated cycles:

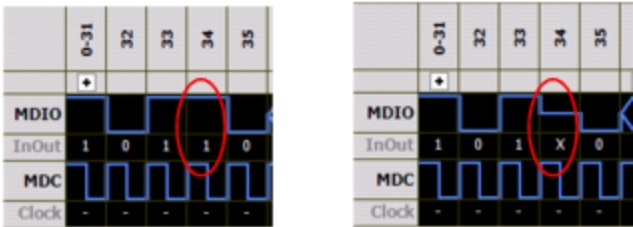


Protocol Definition Editor Window, Repeated Cycles Exposed

Note these points:

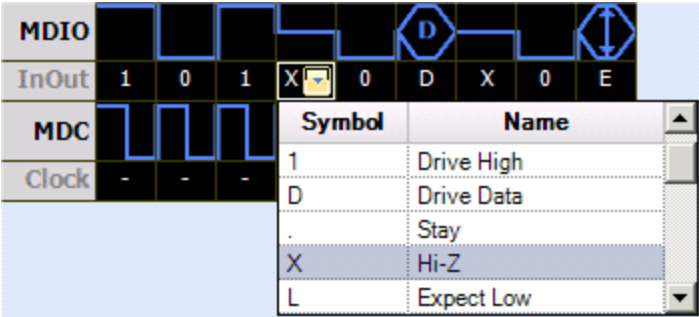
- If a field consists only of repeated cycles, as in this example, you can see the highlighted frame only when the repeated cycles are exposed.
- The numbers of all but the first cycle are grayed out. This indicates that they are repeated cycles.
- You can edit only the first cycle, not the repeated cycles. Editing is described below.

You can edit the bit action character cells in the Frame window. Changing the character immediately changes its graphical representation in the cell above it. In the two figures below, the original value for cycle 34,1, is changed to X, for Hi-Z, and the graphic representation changes as well:



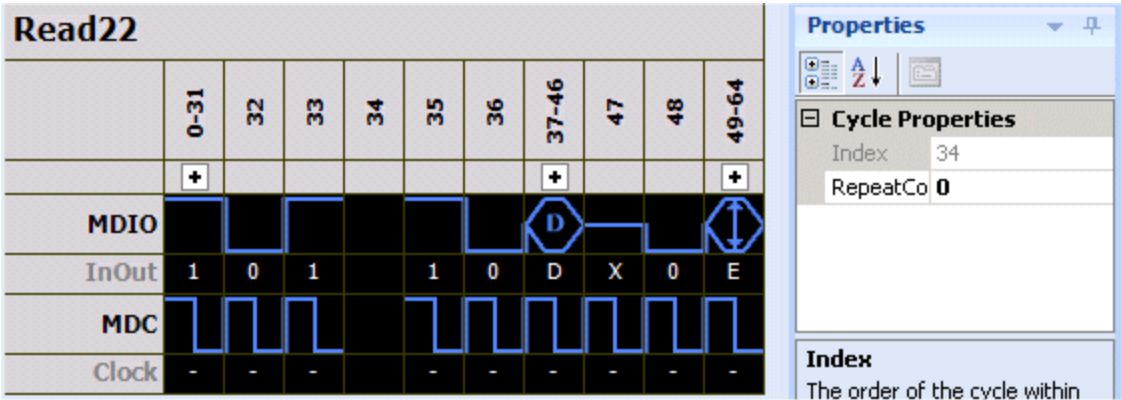
Editing the Value for a Cycle in a Character Cell

You can directly edit a character cell. You are not able to enter an invalid character. For easier entry, when you click the character cell, a two-column list appears with the valid action characters and the name of the bit action that each character maps to:



Editing the Value for a Character Cell Using the List of Valid Characters

You can also use the standard cut, copy, and paste buttons, commands (Edit menu), and control sequences on the character data in the Frame window. You can insert cycles into a specific location in the Frame window. Select the cycle before which you want to insert a new cycle. Then use Insert>Cycle or Ctrl+Y. A new empty cycle is inserted before the selected cycle:



Inserting a Cycle

To make this cycle repeat, edit the Repeat value in the Properties window to indicate how many times the cycle should repeat.

If no cycle is selected in the Frame window, the new cycle is appended as the last cycle.

You can zoom in and out on the Frame window:

- +

Zoom In

Use this button or View>Zoom In or Ctrl+I.
- -

Zoom Out

Use this button or View>Zoom Out or Ctrl+U.

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ProtocolDefinitionEditorUsing.2.07

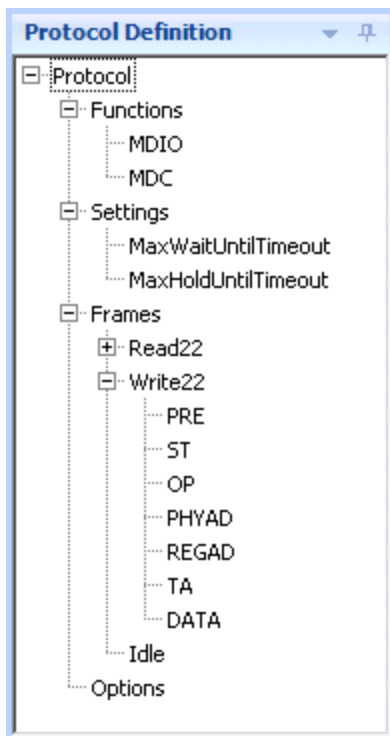


Using the Protocol Definition Editor : Editing a Definition

Editing a Definition

You can edit existing elements and add new elements in the currently loaded protocol definition file.

Note that the section on [Inserting Elements](#) describes additional ways of creating new elements. You can think of a definition file as a set of collections of elements:



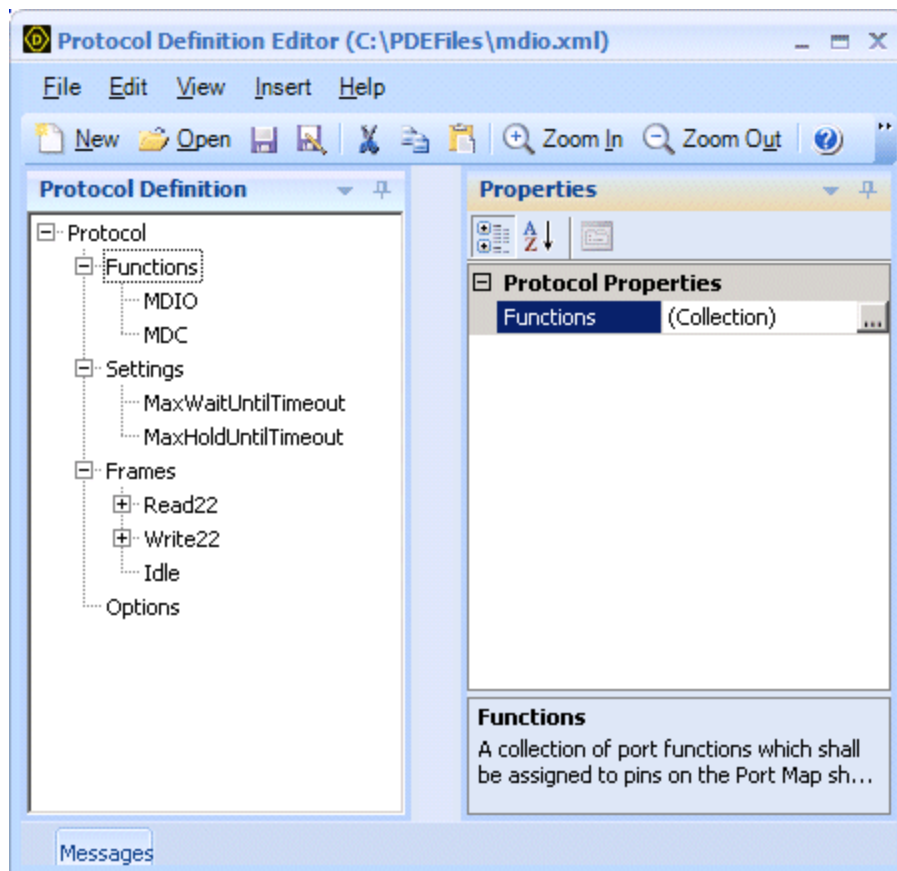
Example Definition File

When you look at the Protocol Definition window, each header type - Functions, Settings, Frames, and Options - is a collection of elements: Settings is a collection of settings, and so on. Frames is a collection of frames, and each individual named frame is a collection of cycles and fields. Each individual named field is a collection of bits.

Protocol Definition Editor provides an editor for each collection. You bring up the collection editor to add new items, remove existing items, and move items within the collection. You can also change the value of the properties of the items in the collection.

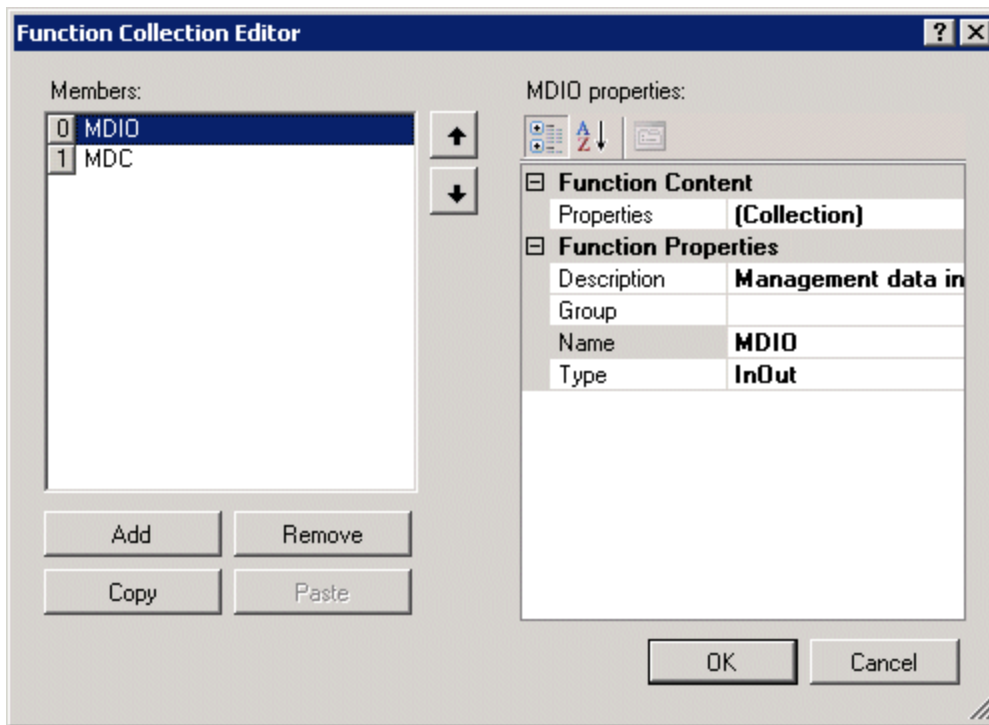
When you select an element in the Protocol Definition window, the Properties window displays its properties and collections. When you select a collection, an ellipsis button (...) appears. Click the button to bring up the collection editor for functions. You can also right-click the item in the Protocol Definition window. The context menu lets you bring up the collections editor for that item. For example, assume you want to add a new function. To add a function, you can do the following:

- In the Protocol Definition window, select Functions. Right-click and select Edit Functions from the context menu.
- With Functions selected in the Protocol Definition window, go to the Properties window. Select Functions, which causes an ellipsis button to appear next to (Collection). Click this button:



Bringing Up the Function Collection Editor

Either of these actions brings up the Function Collection Editor:

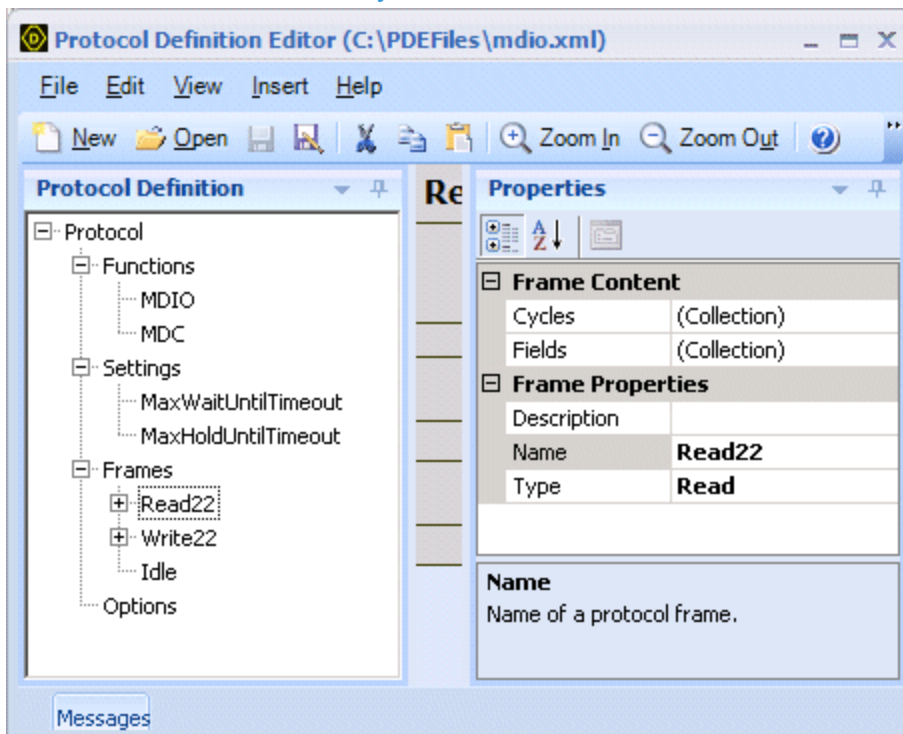


Function Collection Editor

You can add a new function or make a copy of an existing one. You can use the up and down arrows to reorder the functions. You can use the properties on the right side of the editor to specify the name and type of the new function and to add properties, if desired.

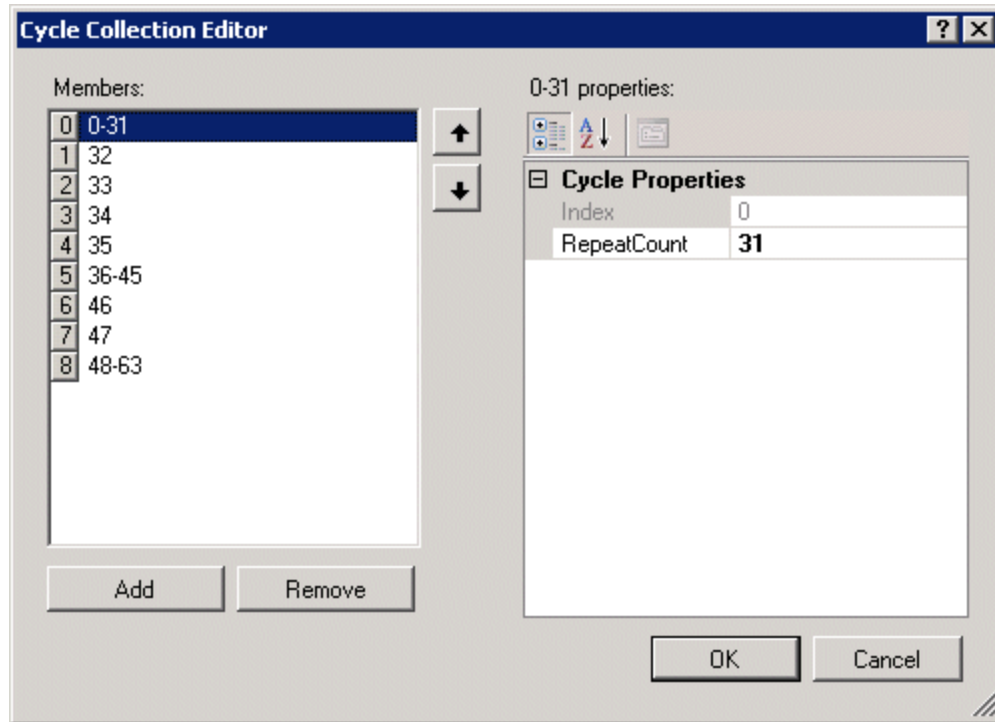
If there is a predefined set of valid values, clicking Add adds one of those values. For example, nWire has two valid settings. Clicking Add adds one setting. Clicking it again adds the other. If you click again, an error is returned saying that all possible settings for the nWire protocol have already been added.

A frame is a collection of cycles and fields:



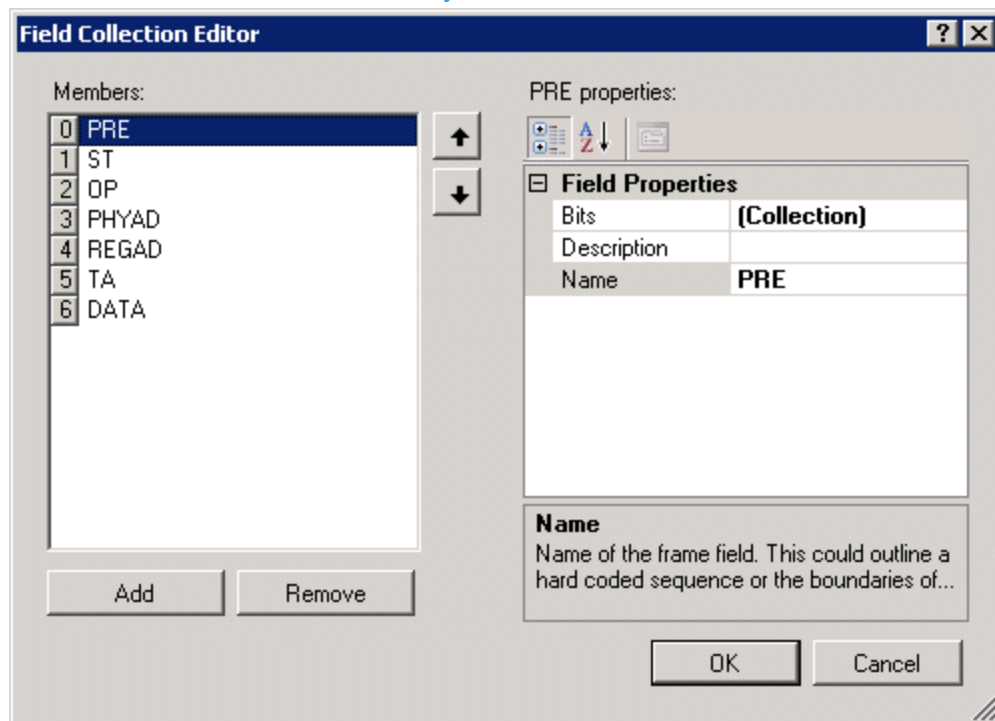
Example Frame Showing Structure and Properties

The Cycle Collection Editor lets you add cycles and add repeat counts to cycles:



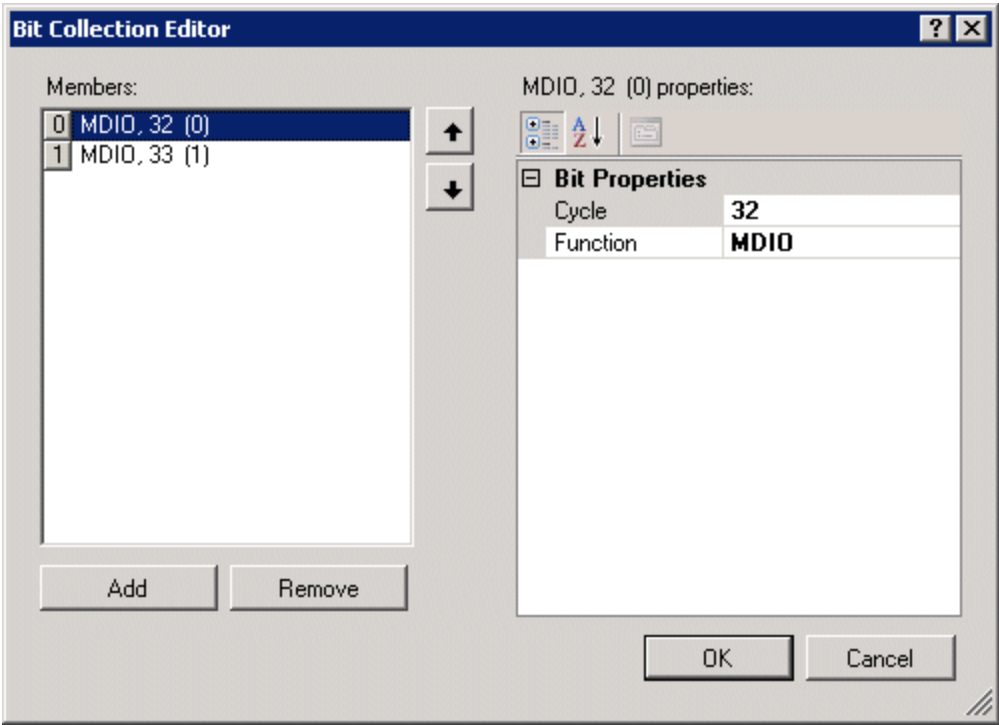
Cycle Collection Editor

The Field Collection Editor lets you add fields:



Field Collection Editor

It also lets you invoke the Bit Collection Editor:



Bit Collection Editor

This collection of bits (function/cycle locations) determines the value of the field: which cycles of which functions comprise the field. The order of the collection is significant: the MSB of the field must be the first bit.

ProtocolDefinitionEditorUsing.2.08

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Using the Protocol Definition Editor : Inserting Elements

Inserting Elements

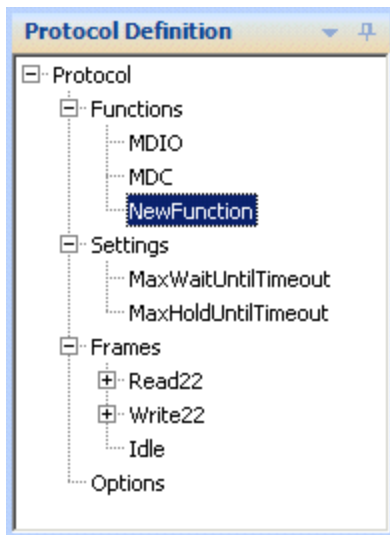
[Editing a Definition](#) describes how to use the Protocol Definition window, Properties window, and collection editors to add new elements to the definition. This section describes another way to insert new elements.

The Insert menu has commands that let you insert new elements. Each command has a corresponding control sequence. The commands are as follows:

- | | |
|---|--------------------------------|
| • | Frame (Ctrl+R) |
|---|--------------------------------|
- | | |
|---|-----------------------------------|
| • | Function (Ctrl+F) |
|---|-----------------------------------|
- | | |
|---|--------------------------------|
| • | Cycle (Ctrl+Y) |
|---|--------------------------------|
- | | |
|---|--------------------------------|
| • | Field (Ctrl+E) |
|---|--------------------------------|
- | | |
|---|----------------------------------|
| • | Setting (Ctrl+T) |
|---|----------------------------------|
- | | |
|---|---------------------------------|
| • | Option (Ctrl+L) |
|---|---------------------------------|

Menu items are enabled when only it is valid to insert the item. For example, Cycle and Field are enabled only when you have selected a frame, because they are inserted into a specific frame.

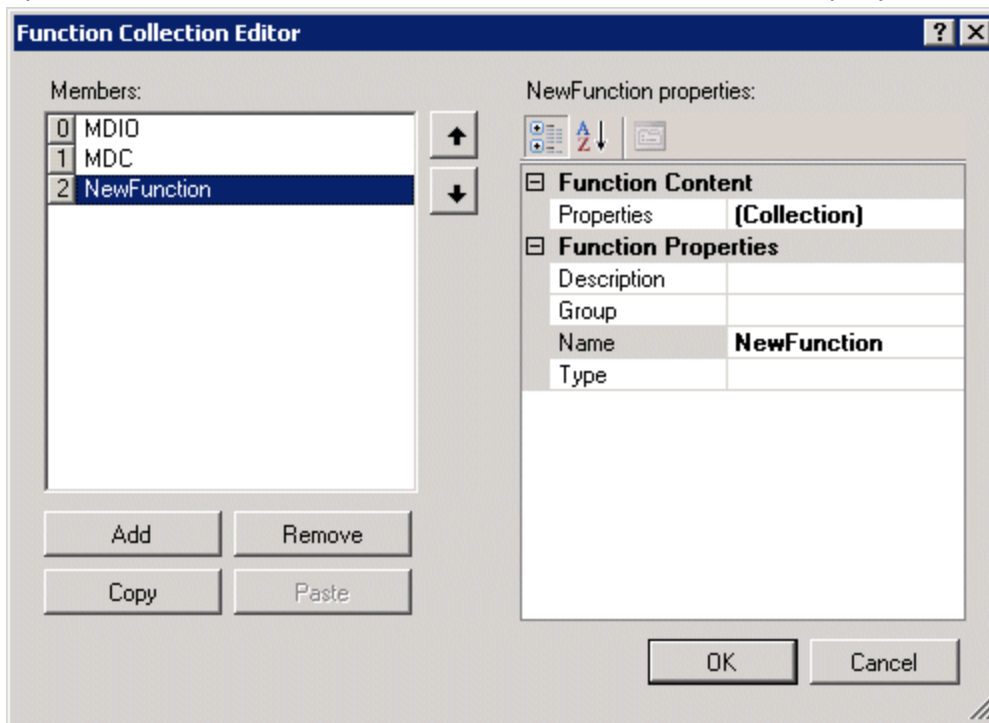
The new item is inserted at the end of the collection:



Inserting a New Item in a Collection

The one exception is if you have selected a cycle in the Frame window. In that case, a new cycle is inserted before the selected one.

Once you insert a new element, you can select it and use the Properties window to rename it and change its properties. You can also right-click the item in the Protocol Definition window and select Edit <item type> (for example, Edit Functions) from the context menu. This lets you move the item up or down in the collection as well as edit the new item's properties:



Editing an Element in the Function Collection Editor

For more information, see [Editing a Definition](#).

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Using the Protocol Definition Editor : Editor Toolbar and Command Reference

Editor Toolbar and Command Reference

This section lists the actions that are available from the Protocol Definition Editor menus and toolbars. References to more complete descriptions are included.

Most toolbar buttons correspond to menu commands, but some are available only on the toolbars, on the menu, or on the right-click context menus.

The section is organized on where the command appears, on the toolbar or on one of the menus:

- [Toolbar](#)
- [File Menu Commands](#)
- [Edit Menu Commands](#)
- [View Menu Commands](#)
- [Insert Menu Commands](#)



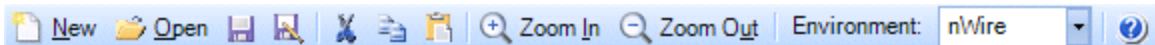
ProtocolDefinitionEditorUsing.2.10



Using the Protocol Definition Editor : Editor Toolbar and Command Reference : Toolbar

Toolbar

The Protocol Definition Editor toolbar looks like this:



Protocol Definition Editor Toolbar

The individual controls are as follows:

	File>New (Ctrl+N) Creates a new protocol definition file. See Creating, Opening, Saving, and Closing Files.
	File>Open (Ctrl+O) Opens an existing protocol definition file. If there are syntax errors in the file, Open returns an error. See Creating, Opening, Saving, and Closing Files.
	File>Save (Ctrl+S) Saves the displayed file to the file you opened. If this is a new file that has not been saved before, it performs a Save As. See Creating, Opening, Saving, and Closing Files.
	File>Save As Opens a browser to let you save the displayed file to a new file you opened. See Creating, Opening, Saving, and Closing Files.
	Edit>Cut (Ctrl+X), Copy (Ctrl+C), Paste (Ctrl+V) Performs standard cut, copy, and paste operations. You can use these on the data in the Frame window. See Frame Window.

 Zoom In

View>Zoom In (Ctrl+I)
Zooms in on the Frame window. See Frame Window.

 Zoom Out

View>Zoom Out (Ctrl+U)
Zooms out on the Frame window. See Frame Window.

Environment:

nWire

Specifies the protocol environment. Currently, only nWire is supported.

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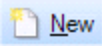
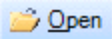
ProtocolDefinitionEditorUsing.2.11



Using the Protocol Definition Editor : Editor Toolbar and Command Reference : File Menu Commands

File Menu Commands

The File menu has commands for opening and saving files. See [Creating, Opening, Saving, and Closing Files](#).

New	 New Ctrl+N Creates a new protocol definition file.
Open	 Open Ctrl+O Opens an existing protocol definition file. If there are syntax errors in the file, Open returns an error.
Save	 Save Ctrl+S Saves the displayed file to the file that you opened. If this is a new file that has not been saved before, it performs a Save As.
Save As	 Opens a browser to let you save the displayed file to a new file that you opened.
Exit	 Exits from the editor. If the currently open file has changes that have not been saved, you are asked if you want to save the changes.

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Using the Protocol Definition Editor : Editor Toolbar and Command Reference : Edit Menu
Commands

Edit Menu Commands

The Edit menu includes standard editing commands:

- | | |
|---|----------------|
| • | Cut (Ctrl+X) |
| • | Copy (Ctrl+C) |
| • | Paste (Ctrl+V) |

 They have the standard corresponding toolbar buttons.
You can use these on the data in the Frame window. See Frame Window.

		
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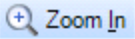
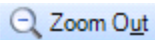
ProtocolDefinitionEditorUsing.2.13



Using the Protocol Definition Editor : Editor Toolbar and Command Reference : View Menu
Commands

View Menu Commands

View menu commands let you zoom in and out on the Frame window:

-  Zoom In or (Ctrl+I)
-  Zoom Out (Ctrl+U)



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Using the Protocol Definition Editor : Editor Toolbar and Command Reference : Insert Menu Commands

Insert Menu Commands

The Insert menu has commands that let you insert new elements. Each command has a corresponding control sequence. The commands are as follows:

- [Frame](#) (Ctrl+R)
- [Function](#) (Ctrl+F)
- [Cycle](#) (Ctrl+Y)
- [Field](#) (Ctrl+E)
- [Setting](#) (Ctrl+T)
- [Option](#) (Ctrl+L)

See [Inserting Elements](#).



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